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ARIA-NBV: Quality-Driven Next-Best-View Planning with Egocentric Foundation Models

*ARIA-NBV: Qualitaetsgetriebene Next-Best-View-Planung mit egozentrischen Foundation-
Modellen*

Master's Thesis Proposal / Proposal zur Masterarbeit

for obtaining the academic degree Master of Science (M.Sc.)

submitted to
Munich University of Applied Sciences
Department of Computer Science & Mathematics

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Program: Informatik - Master
Specialization: Visual Computing and Machine Learning
Matriculation Number: 00807819

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Submission Date: 30 April 2026

Transparency in the Use of AI Tools

AI-assisted tools were used to organize literature notes, check consistency across repository documentation, and draft parts of the proposal text. The author remains responsible for the final research scope, technical claims, citations, and submitted document.

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1 Motivation

Active 3D reconstruction is limited not only by the quality of the reconstruction algorithm, but also by the choice of viewpoints that provide its input. A mobile or wearable camera cannot observe all surfaces at once, and each additional acquisition consumes time, motion budget, and operator attention. The next-best-view (NBV) problem therefore asks which view should be selected next so that a reconstruction improves as much as possible under a constrained acquisition budget. This question is especially important for egocentric indoor capture, where observations are partial, clutter is common, and many useful surfaces are visible only from a narrow range of viewpoints. The active-perception literature makes the same point at a broader level: perception improves when the agent can choose actions that change what is observable [AWB88, Baj88].

Many NBV systems use coverage or information gain as a proxy objective. These objectives are attractive because they can often be computed from occupancy, uncertainty, or visibility fields, but they do not necessarily measure whether the final reconstructed surface becomes more accurate. VIN-NBV introduced a more direct quality-driven objective by learning to rank candidate views according to Relative Reconstruction Improvement (RRI), an oracle signal derived from the reduction in Chamfer-style reconstruction error after adding a candidate observation [Fra+25]. This idea is a strong foundation for the thesis because it separates the desired scientific target, reconstruction-quality improvement, from convenient but weaker proxies such as the number of newly covered voxels.

ARIA-NBV extends this quality-driven view-ranking idea to egocentric indoor data from the Project Aria ecosystem. Aria Synthetic Environments (ASE) provides synthetic egocentric trajectories, calibrated camera streams, semi-dense SLAM points, object annotations, and ground-truth meshes for a supervised subset [Met25]. EFM3D and EVL provide a compatible egocentric foundation-model representation that lifts multi-stream observations into a local 3D voxel field [Str+24a]. This makes it possible to compute expensive oracle RRI labels in simulation, train a lightweight candidate scorer on frozen egocentric features, and keep the data representation close to real Aria recordings.

The thesis is motivated by the gap between one-step candidate ranking and useful multi-step view planning. The current implementation already supports discrete candidate generation, oracle label computation, immutable offline stores, VIN-style scoring, and early counterfactual rollout scaffolding. However, the learned scorer is still preliminary and should not be presented as a finished NBV policy. The next credible step is to make the data and geometry trustworthy, then test whether bounded non-myopic search over trusted candidate views improves cumulative reconstruction quality over one-step greedy selection.

THESIS POSITION

The core claim is deliberately narrow: ARIA-NBV should first prove that trusted, mesh-supervised candidate utility supports better one-step scoring, bounded multi-step selection, and a target-conditioned fitted Q_H model over finite candidates in egocentric snippets. Continuous control, external simulators, and actor-critic policies become credible only after that finite-candidate foundation is reproducible.

The long-term research direction includes target-aware and task-aware view planning. Indoor reconstruction is rarely uniform in importance: a user may care more about an object, doorway, appliance, or region of missing surface than about a large already-visible wall. The thesis therefore proposes target-conditioned, quality-driven NBV as its central direction. It keeps full continuous reinforcement learning, simulator-backed online control, and real-device deployment as later extensions, while focusing the thesis core on target-specific RRI, discrete candidate scoring, bounded rollout evaluation, and fitted Double-Q / Q_H over finite candidate sets.

2 Related Work and Positioning

The thesis sits inside the older active-perception tradition in which the sensing action is part of the perceptual computation rather than a passive data-collection step. Bajcsy’s formulation of active perception and the active-vision analysis of Aloimonos, Weiss, and Bandyopadhyay both frame motion as a way to change the constraints of perception [AWB88, Baj88]. Classical view-planning work then

specialized this idea to 3D acquisition: surveys such as Scott, Roth, and Rivest and object-scanning systems such as Banta and coauthors describe the dominant generate-and-test pattern in which candidate views are sampled, scored, and selected under sensing and registration constraints [Ban+00, SRR03]. ARIA-NBV keeps this discretized candidate-set structure, but replaces generic unknown-area or coverage gain with an oracle reconstruction-quality target.

Modern robotic NBV methods extend the same generate-and-test pattern toward online exploration and computational efficiency. Receding-horizon NBV plans over a sampled tree to explore 3D environments, while shadowcasting and projection-based variants reduce the cost of estimating view utility in large maps or voxel structures [Bat+22, Bir+16, Jia+25]. These works are important because they show that horizon, branch factor, and evaluator cost are first-order design variables. They also clarify why ARIA-NBV should start with bounded candidate rollouts: the expensive part is not choosing an RL algorithm, but making the view utility trustworthy enough that a deeper search tree means something.

Learning-based NBV work has recently split into continuous policy learning and discrete quality-driven ranking. GenNBV demonstrates a generalizable continuous 5-DoF policy trained with PPO-style reinforcement learning and coverage rewards [Che+24, Sch+17]. Hestia improves that line by decomposing continuous control into a look-at proposal and a feasible camera pose while using directional voxel-face visibility and close-greedy rewards [Lu+26]. VIN-NBV instead remains closer to candidate ranking: it samples views, computes oracle RRI, and learns to rank candidates by expected reconstruction improvement [Fra+25]. The thesis follows VIN-NBV for the first contribution because RRI is aligned with mesh-supervised reconstruction quality, then uses GenNBV and Hestia as controlled references for later continuous or hierarchical extensions.

Radiance-field and Gaussian-splatting work broadens the representation side of the literature. NeRF and 3D Gaussian Splatting made differentiable or explicit radiance-field representations central to view synthesis and reconstruction [Ker+23, Mil+20]. ActiveNeRF, FisherRF, Next Best Sense, and recent semantic, dynamic, and object-centric 3DGS NBV papers select views by uncertainty, Fisher information, depth uncertainty, semantics, dynamics, or object-specific utility [Jeo+26,

JLD24, LJD25, Pan+22, Str+24b]. These papers are not direct replacements for ARIA-NBV’s oracle labels, but they support the target-aware thesis direction: utility should be decomposed by geometry, object, and task rather than collapsed into undifferentiated scene coverage.

The reinforcement-learning literature enters the thesis as a finite-candidate planning scaffold, not as a commitment to immediate actor-critic control. Trajectory Transformer shows that offline control can be decoded as a sequence problem with beam search, and Gumbel-Top-k provides a stochastic top-k mechanism for sampling diverse high-scoring sequences without replacement [JLL21, KHW19]. Double DQN supplies the first mandatory learned value method for ARIA-NBV: a masked fitted Double-Q / Q_H model over finite candidate sets, trained from ASE oracle rollout data [HGS15]. IQL, deep energy-based policies, PPO, and SAC remain gated follow-up or bridge references because they expose support, reward-loop, and simulator assumptions that the thesis core should not hide [Haa+17, Haa+18, KNL21, Sch+17]. ARIA-NBV therefore treats **ArgTopK -> ArgTop1_1 -> ... -> ArgTop1_H -> Q_H** as the thesis-grade non-myopic ladder before optional actor-critic or continuous-control work.

Literature family	Main lesson	ARIA-NBV boundary
Active perception and classical view planning	Motion and viewpoint choice are part of perception; candidate views are usually generated and evaluated under constraints.	Keep typed candidate sets and explicit geometry contracts, but use reconstruction-quality labels.
Continuous and hierarchical NBV policies	Learned policies can operate in 5-DoF spaces when simulator, reward, and state are mature.	Use as extension references after discrete RRI rollouts show headroom.
Radiance-field and 3DGS active selection	Uncertainty, Fisher information, semantics, dynamics, and object utility are separable channels.	Borrow target-aware decomposition, not the objective, until calibrated against RRI.
Offline and stochastic planning	Beam search, Gumbel-Top-k, offline value learning, and maximum-entropy policies define later baselines.	Begin with deterministic bounded oracle rollout, train masked fitted Double-Q / Q_H after rollout data are trusted, and gate continuous RL behind evidence.

Table 1: Literature positioning for the thesis scope.

3 Problem Statement

The thesis studies view selection for an egocentric reconstruction episode in which the agent has already observed a partial trajectory and must choose future camera views from a finite candidate set. At step t , the available state consists of the current reference pose, calibrated camera geometry, accumulated semi-dense points, candidate poses, candidate validity information, and frozen egocentric features derived from the logged snippet. The privileged ground-truth mesh is used only to compute oracle labels and evaluation metrics. It is not an actor-visible input for the first planning formulation.

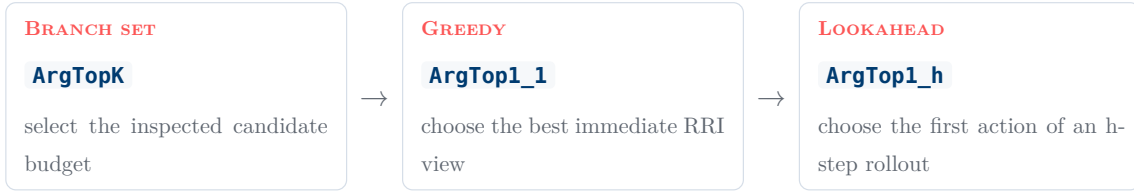
The core quality signal is RRI. Given the current point set, a candidate view, and the ground-truth mesh, the oracle renders candidate geometry, fuses it with the current reconstruction state, and measures the relative reduction in reconstruction error. In the scene-level case this score evaluates improvement over the full mesh, while in the target-aware case it evaluates the same improvement over an object or region crop. The thesis keeps these two quantities visible rather than hiding the tradeoff inside a single opaque reward.

$$\text{RRI}(q) = \frac{\text{CD}(\mathcal{P}_t, \mathcal{M}_{\text{GT}}) - \text{CD}(\mathcal{P}_t \cup \mathcal{P}_q, \mathcal{M}_{\text{GT}})}{\text{CD}(\mathcal{P}_t, \mathcal{M}_{\text{GT}}) + \varepsilon}$$

The central research problem is to learn and evaluate a target-conditioned scorer that ranks candidate views by target-specific reconstruction improvement, then train a finite-candidate fitted Q_H selector from trusted rollout data. The one-step baseline selects the candidate with the best immediate oracle or predicted RRI. The non-myopic baseline restricts the search tree to a small set of high-scoring valid candidates and recursively evaluates the first action under a finite horizon. This produces the experimental sequence

$$\text{ArgTopK} \rightarrow \text{ArgTop1}_1 \rightarrow \text{ArgTop1}_2 \rightarrow \dots \rightarrow \text{ArgTop1}_H$$

, where ArgTop1_1 is the greedy selector and ArgTop1_H is the first action chosen by an H -step bounded rollout. The learned Q_H model is then evaluated against one-step greedy and model scoring under the same acquisition budget.



The proposal deliberately narrows the problem to discrete candidate sets. Continuous policies such as GenNBV demonstrate that direct 5-DoF reinforcement learning is possible when a simulator, reward, and training distribution are available [Che+24]. Hestia further shows that hierarchical action factorization and close-greedy reward design can improve continuous NBV control [Lu+26]. In ARIA-NBV, however, oracle RRI evaluation is expensive, counterfactual modalities are incomplete, and the local offline store must first be made inspectable. The thesis therefore treats finite-candidate fitted Double-Q / Q_H as the required value-learning result and continuous actor-critic learning as a follow-up direction rather than as the first claim.

The main failure mode the thesis must avoid is a visually plausible but geometrically inconsistent pipeline. Candidate frusta, camera frames, display rotations, invalid masks, and RRI labels can all look reasonable while encoding the wrong transformation or supervision contract. For this reason, the proposal treats offline-store validation and Rerun-based inspection as part of the scientific method rather than as optional tooling. If a sample lacks required pose, camera, point, or metric fields, it should fail before rendering or training; if optional fields are absent, the diagnostics should state that limitation explicitly.

4 Objectives

The thesis objective is to produce a reproducible target-aware NBV stack for egocentric indoor reconstruction. The expected outcome is not a claim of finished continuous-control reinforcement learning. It is a controlled study of whether target-specific RRI supervision, learned candidate scoring, bounded multi-step rollout, and finite-candidate fitted Double-Q / Q_H selection can improve reconstruction quality under the same view budget as one-step greedy selection.

The primary research question is whether a target-conditioned NBV model can produce acquisition-cost-efficient trajectories that maximize target-specific RRI

for a selected target of interest. This question decomposes into five measurable objectives. First, the data and oracle contracts must be trustworthy enough that candidate geometry and RRI labels can be inspected sample by sample. Second, target-specific RRI must be defined using ground-truth oriented bounding box crops before predicted targets or open-vocabulary targets are introduced. Third, a VIN-style scorer must rank candidates by scene-level and target-level RRI on held-out snippets. Fourth, bounded multi-step rollout must be compared against one-step greedy selection under equal candidate and acquisition budgets. Fifth, a target-conditioned fitted Q_H model must learn bounded cumulative target RRI from the trusted rollout traces and beat one-step greedy or model scoring under the same acquisition budget, or produce a documented blocker.

Objective	Evidence	Exit condition
Offline trust contract	Immutable VIN offline samples expose required pose, camera, point, candidate, and RRI fields.	One train and one validation sample load without online fallback and can be inspected geometrically.
Target-specific RRI	Ground-truth OBB crops define target mesh and target point subsets for oracle evaluation.	Target RRI and scene RRI are reported side by side on a trusted subset.
Target-conditioned scorer	A VIN-style model receives a target encoding and ranks candidates by target-specific RRI.	Held-out rank correlation, top-k oracle hit rate, and calibration are compared against a scene-level scorer.
Bounded rollout	Greedy, random, bounded oracle rollout, stochastic or beam rollout, model-scored rollout, and fitted Q_H are evaluated with the same budget.	Q_H improves cumulative target RRI over one-step greedy/model scoring or the limitation is documented with failure cases.

Table 2: Planned thesis objectives and evidence gates.

The required contributions are the oracle and data contract, the target-specific RRI formulation, the target-conditioned candidate scorer, bounded rollout evaluation, and masked fitted Double-Q / Q_H over finite candidate sets. Optional extensions include richer entity tokens, open-vocabulary target selection, IQL-style offline value learning after Q_H is stable, and continuous target-then-pose control. These extensions become thesis-core only if the earlier evidence gates show that the finite-candidate target-aware setting is reliable and time remains.

The thesis will evaluate success using reconstruction-quality and planning metrics rather than only training loss. The core metrics are target RRI, scene RRI, cumulative target RRI, acquisition cost, invalid-action rate, runtime, rank correlation, top-k oracle hit rate, ordinal calibration, and representative failure visualizations. This metric set keeps model quality, planning quality, and data validity visible as separate axes.

5 Proposed Method

The proposed method follows a staged diagnostics-first pipeline. The first stage fixes the data substrate. ARIA-NBV reads fixed-length ASE snippets through the active data-handling path, builds immutable VIN offline stores for expensive oracle products, and validates that train and validation samples expose the fields needed for reconstruction-quality supervision. The sample contract includes scene and snippet metadata, reference pose, candidate poses, camera intrinsics or PyTorch3D cameras, semi-dense points, candidate validity information, oracle RRI, accuracy and completeness components, and optional diagnostics such as candidate point clouds, depth maps, meshes, object boxes, and voxel bounds.

The second stage computes scene-level and target-level oracle labels. Scene-level RRI follows VIN-NBV by evaluating how much a candidate view reduces the Chamfer-style distance between the current reconstruction and the ground-truth mesh [Fra+25]. Target-level RRI uses the same metric but restricts the evaluation to the selected target crop. The first target definition is a ground-truth OBB crop because it is inspectable and directly available in the ASE/EFM3D data. Predicted boxes, target tokens, or open-vocabulary selection are treated as realism ablations after the ground-truth target oracle is trusted.

$$\text{RRI}_{\text{total}(q)} = \sum_{e \in \mathcal{E}} \mathbf{w}_e \cdot \text{RRI}_e + \lambda_{\text{scene}} \cdot \text{RRI}$$

The third stage trains and analyzes a VIN-style candidate scorer. The current implementation already uses frozen EVL/EFM3D features, candidate-relative pose information, semi-dense projection cues, and an ordinal prediction head. The thesis will keep this architecture bounded and evidence-driven. It will first establish the reproducible one-step scene-level baseline, then add target conditioning

through explicit target fields. The model will be judged by ranking and calibration metrics on held-out snippets, not by loss alone. This is important because NBV acts through ordering: a model that predicts approximate scalar values but ranks candidates poorly is not useful for view selection.

The fourth stage evaluates bounded non-myopic planning. For a state s_t and valid candidate set $C(s_t)$, the one-step oracle baseline selects the best immediate candidate. A bounded rollout first chooses the top K candidates under the one-step score and searches only those branches to horizon H . This is closer to receding-horizon and projection-efficient NBV planning than to unrestricted policy learning: horizon and branch factor are explicit compute budgets, and every transition can be inspected [Bir+16, Jia+25]. The value recursion is

$$V_{h(s_t)} = \max_{q \in \text{ArgTopK}(s_t)} (r(s_t, q) + V_{h-1}(T(s_t, q)))$$

and the selected action is

$$\text{ArgTop1}_{h(s_t)} = \underset{q \in \text{ArgTopK}(s_t)}{\text{argmax}} (r(s_t, q) + V_{h-1}(T(s_t, q)))$$

This formulation imports the useful part of sequence-model planning without immediately training a trajectory model. Trajectory Transformer shows that offline control can be treated as sequence decoding with beam search [JLL21], while Gumbel-Top-k provides a principled stochastic beam extension for diverse samples without replacement [KHW19]. ARIA-NBV will first use deterministic bounded rollout because it is easier to reproduce and inspect. Stochastic beams diversify the rollout data after deterministic traces are trusted.

The fifth stage gates and then trains the required finite-candidate value model. If the M1 data/oracle contract is passable, target-aware labels are trusted, deterministic rollouts show headroom, and the VIN scorer ranks held-out candidates reliably, ARIA-NBV trains a target-conditioned fitted Double-Q / Q_H model on masked finite candidate sets. Double Q-learning is the first value method because selector/evaluator separation reduces overestimation in max-based targets [HGS15]. Q_H predicts bounded cumulative target RRI and must beat one-step greedy/model scoring under the same acquisition budget or produce an explicit

blocker. IQL is a second offline-RL ablation only after Q_H is stable [KNL21]. PPO and SAC remain later references for simulator-backed continuous control rather than first thesis baselines, because they assume a reward loop and interaction regime that the current expensive oracle does not yet provide [Haa+18, Sch+17]. These methods will not be used to claim unrestricted continuous control unless the data support, reward speed, simulator access, and evaluation protocol are all sufficient.

Diagnostics are part of the method. A Rerun offline inspector will render semi-dense points, reference poses, candidate frusta, optional meshes, target boxes, and RRI-colored candidate layers from the immutable offline store. Candidate frusta should default to batched manual line strips until pose and camera regression tests prove that native camera logging is frame-safe. This inspector is intended to catch frame mistakes, invalid candidates, label pathologies, and missing fields before they contaminate training or rollout results.

6 Schedule

The planned thesis window runs from 29 April 2026 to 30 September 2026. The schedule mirrors the current roadmap and keeps the highest-risk dependencies early: offline-store trust, geometry inspection, and oracle correctness precede target-conditioned modeling and rollout claims. Writing and figure preparation run throughout the project, but the final thesis narrative is frozen only after the August evidence gate.

Dates	Milestone	Exit condition
29 Apr–10 May	Scope and diagnostics foundation	The worktree is classified, the partial offline-store smoke path is understood, and the Rerun inspector workflow is documented.
11 May–31 May	Data, cache, and oracle correctness	Offline-store contracts, split handling, frame conventions, candidate validity, and oracle throughput are stable enough for model claims.
1 Jun–21 Jun	VIN baseline and calibration	A reproducible one-step VIN baseline is trained and evaluated with rank, calibration, ordinal-bin, and top-k diagnostics.
22 Jun–12 Jul	Target-specific RRI	Ground-truth OBB-cropped target RRI is implemented, inspected, and compared with scene-level RRI on a trusted subset.

Table 3: Planned thesis schedule through the target-specific oracle milestone.

Dates	Milestone	Exit condition
13 Jul–9 Aug	Target-conditioned VIN	The model accepts a target encoding and is evaluated against scene-level, target-level, oracle, and geometric baselines.
10 Aug–30 Aug	Multi-step rollouts and Q_H	Greedy, random, bounded oracle, stochastic or beam, model-scored, and fitted Q_H selections are compared by cumulative target RRI and acquisition cost.
31 Aug–13 Sep	Continuous bridge and optional IQL	IQL or actor-critic bridge design is attempted only after fitted Q_H evidence is stable; no quantitative continuous-control baseline is required.
14 Sep–30 Sep	Thesis freeze and release	Final figures, ablations, failure cases, reproducible configs, and demonstration smokes are complete.

Table 4: Planned thesis schedule from target-conditioned modeling to release freeze.

The main risk is that oracle or offline-store trust takes longer than expected. In that case, the thesis remains valid as a stronger geometry-first study of scene-level RRI, target-specific RRI, and one-step candidate scoring, while learned multi-step control is reduced to oracle rollout analysis. A second risk is that target-specific labels are too sparse or unstable for reliable model training. The fallback is to report target RRI as an oracle evaluation and diagnostic tool while keeping the learned model scene-level. A third risk is that bounded lookahead does not improve over one-step greedy under the available candidate budget. That outcome is still

scientifically useful if it is paired with failure analysis, because it identifies whether candidate generation, target observability, or the reward definition is the limiting factor.

The mandatory value-learning path is fitted Double-Q / Q_H over finite candidate sets. It will be pursued only after held-out VIN ranking, oracle-evaluated VIN-selected rollouts, calibration and stage-shift analysis, and Rerun failure visualization are available. If those gates are not met by the end of August, the thesis will document the blocker explicitly rather than reclassifying Q_H as optional. The stretch path after a stable Q_H result is IQL, actor-critic bridge design, or continuous target-then-pose control; if the gates are weak, the thesis will prioritize robust target-aware ablations and a defensible future-work discussion over a weak continuous-RL claim.

7 Preliminary Thesis Outline

The thesis will be written as an empirical methods thesis. It starts from the quality-driven NBV problem, introduces the Aria data and geometry contracts, then develops the oracle, model, and rollout components in the same order in which they are validated. The preliminary bibliography is rendered at the end of this proposal from the shared repository bibliography.

Chapter	Purpose	Expected evidence
Introduction	Motivate target-conditioned, quality-driven NBV for egocentric indoor reconstruction and state the thesis questions.	Problem definition, scope boundary, and contribution summary.
Background and related work	Review active perception, classical view planning, NBV, RRI, egocentric foundation models, target-aware reconstruction, bounded rollout, and offline value learning.	Literature synthesis centered on VIN-NBV, EFM3D, GenNBV, Hestia, radiance-field NBV, Trajectory Transformer, Gumbel-Top-k, Double Q-learning, and IQL.
Data and geometry contracts	Describe ASE snippets, camera and pose frames, immutable offline stores, candidate generation, and Rerun inspection.	Offline-store inventory, frame-safety checks, and visual diagnostics.
Oracle RRI and target-specific RRI	Define scene-level and target-level reconstruction-quality labels, invalid cases, and accuracy/completeness components.	Trusted oracle examples, label distributions, target crops, and throughput measurements.
Target-conditioned candidate scoring	Present the VIN-style scorer, target encoding, ordinal objective, and calibration analysis.	Held-out ranking, top-k oracle hit rate, calibration, ablations, and failure cases.
Bounded rollout and Q_H evaluation	Compare one-step greedy, random, oracle rollout, stochastic or beam rollout, model-scored rollout, and fitted Double-Q / Q_H under equal budget.	Cumulative target RRI, scene RRI, Q_H success bar, acquisition cost, invalid-action rate, runtime, and trajectory visualizations.
Discussion and conclusion	Interpret limits, failure modes, simulator and deployment gaps, and future continuous-control or entity-aware extensions.	Evidence-gated conclusion and reproducibility notes.

Table 5: Preliminary thesis chapter outline.

The expected final contribution is a reproducible target-aware finite-candidate view-selection study, not a broad claim that continuous reinforcement learning has been solved for egocentric reconstruction. The thesis will therefore separate implemented results from extensions. The required result includes fitted Double-Q / Q_H over finite candidate sets if the prerequisite gates pass; SceneScript-style semantic memory, 3D Gaussian Splatting simulators, open-vocabulary targets, actor-critic control, and real-device guidance are future directions unless they become necessary to explain the core target-aware RRI and Q_H results.

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